

UI Design Patterns

UI design patterns—such as the Form Validation selected for the "Appointment Registration" task—act as a shared language between the system and the user. By utilizing recurring components, we respect the mental models that UADY staff already possess from their experience with other digital tools.

We aren't reinventing the wheel; we're smoothing the path. In a medical context, where time is a critical resource and cognitive load is high, forcing a user to "decipher" how to fill out a form is a design flaw that can lead to actual data entry errors.

Implementing robust form validation on the frontend (instead of relying solely on the Django backend) is an act of **technical empathy**.

- **Error as a Guide:** When the system flags an error in real-time, it isn't "scolding" the administrator; it's saving them the step of submitting, waiting for a server response, and starting over.
- **Security and Trust:** A system that validates data correctly projects professionalism. For a doctor who relies on the schedule to organize their day, knowing the system prevents duplicate appointments or incomplete records creates indispensable psychological safety in their workflow.

In development, we often focus on technical metrics (like cyclomatic complexity or database query efficiency). However, analyzing this repository through the lens of **Human-Computer Interaction** reminds us that true software quality includes usability.

A medical system can have an impeccable Python architecture, but if the design pattern chosen for critical tasks is inadequate, the system will fail its primary purpose: serving people. Selecting a toolkit or style guide must be a conscious process where we verify that these basic patterns are present to ensure the technology becomes invisible, leaving the doctor's task as the only thing that matters.

The UADY Medical System represents a bridge between HCI academic theory and technical practice. By applying validated UI patterns, we stop building "screens" and start building fluid experiences. The specific implementation of form validation is the first step in ensuring this system is not just a database, but a reliable ally in university health management.

References

[1] Interaction Design Foundation, "UI Design Patterns," *IxDF Literature*, 2024. [En línea]. Disponible en: <https://ixdf.org/literature/topics/ui-design-patterns>. [Accedido: 25-mar-2026].

[2] J. Nielsen, "10 Usability Heuristics for User Interface Design," *Nielsen Norman Group*, 15-nov-2020. [En línea]. Disponible en: <https://www.nngroup.com/articles/ten-usability-heuristics/>. [Accedido: 25-mar-2026].

[3] ISO, "Ergonomics of human-system interaction — Part 210: Human-centred design for interactive systems," *ISO 9241-210:2019*, 2019. [En línea]. Disponible en: <https://www.iso.org/standard/77520.html>. [Accedido: 25-mar-2026].

[4] S. Hooper y E. Berkman, *Designing Mobile Interfaces*. Sebastopol, CA, EE. UU.: O'Reilly Media, 2011.